-- Show me the top 5 days when Nike stock price was highest”

```
SELECT Date, Close
FROM nke
ORDER BY Close DESC
LIMIT 5;
```

-- The highest stock prices occurred in November 2021, where Nike experienced a sustained peak period with consistently high values.

Date	Close
11/5/2021	166.2468109
11/4/2021	164.3081665
11/19/2021	163.7836761
11/22/2021	163.1842957
11/9/2021	162.7909546

2. -- “How did Nike stock perform over the years?”

```
SELECT YEAR(STR_TO_DATE(Date, '%m/%d/%Y')) AS Year,
    AVG(Close) AS Avg_Price
FROM nke
GROUP BY Year
ORDER BY Year;
```

-- Nike stock experienced steady long-term growth, reaching its peak around 2021, followed by a decline indicating a market correction phase.

Year	Avg_Price
2000	3.8565405777738113
2001	4.392141723838708
2002	4.795703091337302
2003	5.215412701869045
2004	7.227713130781754
2005	8.048231066206345
2006	8.22932954043825
2007	11.024058934840637
2008	11.851865304596842
2009	11.02285467089285
2010	15.249193694960326

Result 9

3 -- “How much did the stock change day-to-day (profit/loss)?”

```
SELECT
   STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
   Close,
   (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
   / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return
FROM nke;
```

-- The daily returns show that Nike stock experiences both gains and losses, with most changes being small, indicating overall stability with occasional fluctuations.

2000-01-03	4.471373558	NULL
2000-01-04	4.227483273	-0.05454482427746199
2000-01-05	4.471373558	0.05769160260376978
2000-01-06	4.448146343	-0.005194648735720656
2000-01-07	4.448146343	0
2000-01-10	4.523636341	0.016971113848085737
2000-01-11	4.750109673	0.05006444261386756
2000-01-12	4.5875144	-0.03422979345597101
2000-01-13	4.645586014	0.0126586227173478
2000-01-14	4.767531872	0.026249833203497355
2000-01-18	4.703653812	-0.013398559614285972

Result 10

-- “Which days had HIGH volatility (big price changes)?”

```
SELECT *
```

```
FROM (
   SELECT
      STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
      Close,
      (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
      / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return
   FROM nke
) t
WHERE ABS(Daily_Return) > 0.05;
-- High volatility days indicate periods where the stock experienced significant price changes,
suggesting market uncertainty, major events, or strong investor reactions.
```

Date	Close	Daily_Return
2000-01-04	4.227483273	-0.05454482427746199
2000-01-05	4.471373558	0.05769160260376978
2000-01-11	4.750109673	0.05006444261386756
2000-02-08	3.437733412	-0.18231989583313876
2000-02-09	3.135770082	-0.08783791347692785
2000-02-15	2.82799983	-0.05802712946982378
2000-03-13	2.671211004	0.0697675192742481
2000-03-15	3.066085339	0.14285676228823319
2000-03-16	3.327642679	0.08530660796457383
2000-03-27	3.345095873	0.06088544624009873
2000-03-29	3.630157471	0.059423005579790834

Result 11

```
-- Which year was the most risky (highest volatility)?”
SELECT
   YEAR(Date) AS Year,
   STDDEV(Daily_Return) AS Volatility
FROM (
   SELECT
      STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
      (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
      / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return
   FROM nke
) t
GROUP BY Year
```

```
ORDER BY Volatility DESC;
-- The volatility analysis shows that Nike stock experienced the highest fluctuations in the early 2000s and during major global events such as the 2008 financial crisis and 2020 pandemic. Lower volatility in other years indicates relatively stable market conditions.
```

100% 268:52

Result Grid Filter Rows: Search Export:

Year	Volatility
2000	0.03534180662404818
2008	0.03114900063347015
2022	0.026577470147796368
2025	0.026435645764507878
2020	0.026084702503224828
2001	0.02578867759723431
2009	0.02260831647549142
2002	0.0220867718179666
2024	0.020113545329000213
2011	0.01923541090394352
2018	0.018467126138683107

```
-- If I invested 1000 dollars in 2010, what would it be today?
SELECT
   STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
   Close
FROM nke
WHERE YEAR(STR_TO_DATE(Date, '%m/%d/%Y')) = 2010
ORDER BY Date
LIMIT 1;
SELECT
   STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
   Close
FROM nke
ORDER BY STR_TO_DATE(Date, '%m/%d/%Y') DESC
LIMIT 1;
SELECT
   1000 AS Initial_Investment,

   (SELECT Close
    FROM nke
    WHERE YEAR(STR_TO_DATE(Date, '%m/%d/%Y')) = 2010
```

```
    ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')
    LIMIT 1) AS Buy_Price,

   (SELECT Close
    FROM nke
    ORDER BY STR_TO_DATE(Date, '%m/%d/%Y') DESC
    LIMIT 1) AS Current_Price,

   ( (SELECT Close
       FROM nke
       ORDER BY STR_TO_DATE(Date, '%m/%d/%Y') DESC
       LIMIT 1)
     /
     (SELECT Close
       FROM nke
       WHERE YEAR(STR_TO_DATE(Date, '%m/%d/%Y')) = 2010
       ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')
       LIMIT 1)
   ) * 1000 AS Final_Value;
```

-- If an investor had invested 1000 dollars in Nike stock in 2010, the investment would have grown to approximately 4659 dollars, demonstrating strong long-term growth and high return potential.

Initial_Investme...	Buy_Price	Current_Price	Final_Value
1000	13.26636219	61.81000137	4659.152259282619

2. -- Which year was BEST to invest?”

```
   SELECT
   YEAR(STR_TO_DATE(Date, '%m/%d/%Y')) AS Year,
   MIN(Close) AS Lowest_Price,
   MAX(Close) AS Highest_Price,
   (MAX(Close) - MIN(Close)) / MIN(Close) * 100 AS Return_Percentage
FROM nke
GROUP BY Year
ORDER BY Return_Percentage DESC;
```

-- The analysis shows that the year 2020 provided the highest return, indicating that periods of high volatility can create strong investment opportunities. Not all years yield equal returns, highlighting the importance of timing in stock market investments.

Year	Lowest_Price	Highest_Price	Return_Percentage
2020	58.0899086	134.1157684	130.87619111867565
2000	2.467966795	5.305778503	114.98581398053209
2022	78.58255768	156.1124725	98.66046246002006
2009	7.727622509	13.44094563	73.93377606561347
2001	3.389112234	5.557043076	63.9675139775852
2003	4.006438732	6.504787445	62.35834066412473
2002	3.709376574	6.021103382	62.32116804218539
2008	8.644319534	13.83758068	60.07715385316067
2013	21.99991226	34.24993515	55.68214429778821
2010	12.44419575	19.02050018	52.8463595568239
2025	52.31077194	79.84435272	52.63462908094871

```
-- “Which month gives the best return (best time to invest)?”
SELECT
   MONTH(STR_TO_DATE(Date, '%m/%d/%Y')) AS Month,
   MIN(Close) AS Lowest_Price,
   MAX(Close) AS Highest_Price,
   (MAX(Close) - MIN(Close)) / MIN(Close) * 100 AS Return_Percentage
FROM nke
GROUP BY Month
ORDER BY Return_Percentage DESC;
-- The analysis shows that months like February and March offer the highest return potential,
indicating possible seasonal trends in stock performance. However, these values represent
maximum potential returns and should be interpreted with consideration of market volatility.
```

Month	Lowest_Price	Highest_Price	Return_Percentage
2	2.467966795	139.5245361	5553.420312731558
3	2.485387802	135.2469177	5341.682685944075
10	3.396310329	156.6752625	4513.102082050642
9	3.41434288	154.1184998	4413.855380570331
11	3.723104	166.2468109	4365.274429615719
6	3.280330896	145.8141937	4345.106250646977
8	3.689644098	162.5508118	4305.595973013004
7	3.752279282	156.6228638	4074.072664349189
12	3.886501789	160.7097931	4035.0757525664426
1	4.10092926	156.1124725	3706.758483319949
4	3.460285425	128.8781433	3624.494585587546

```
-- Which month gives CONSISTENT returns?”
SELECT
   MONTH(Date) AS Month,
   AVG(Daily_Return) AS Avg_Return,
   STDDEV(Daily_Return) AS Volatility
FROM (
   SELECT
      STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
      (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
      / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return
   FROM nke
) t
GROUP BY Month
ORDER BY Avg_Return DESC;
```

-- The analysis shows that November provides the most consistent and highest returns, making it the best month for investment. July offers the lowest volatility, indicating stability, while January and February show negative average returns, suggesting weaker performance during the start of the year.

Month	Avg_Return	Volatility
9	0.0014694335618050803	0.022139186841250748
12	0.0012134301621722163	0.02012339790647175
7	0.0009717113632734195	0.014822810970833477
3	0.0008800026612461917	0.024384098957386347
10	0.0008468888744965717	0.019302089598777723
6	0.0002794733140975257	0.02399969392950009
5	0.00026671122082420873	0.017579779811671244
4	0.0002641550808007141	0.018773536139468264
8	0.00007129623354129297	0.016940071880453843
1	-0.0004986900261138 32	0.016094121102869303
2	-0.0006897418256015964	0.02089547685025825

```
-- “What is the longest winning streak (continuous profit days)?”
SELECT
   STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
   Close,

   (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
   / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return

FROM nke;
SELECT
   Date,
   Close,
   Daily_Return,

   CASE
      WHEN Daily_Return > 0 THEN 1
      ELSE 0
   END AS Profit_Day

FROM ( ... ) t;
SELECT
   Date,
   Profit_Day,

   ROW_NUMBER() OVER (ORDER BY Date) -
   ROW_NUMBER() OVER (PARTITION BY Profit_Day ORDER BY Date) AS grp

FROM ( ... ) t;
```

```
SELECT
   Profit_Day,
   COUNT(*) AS Streak_Length
FROM (
   SELECT
      Profit_Day,
      ROW_NUMBER() OVER (ORDER BY Date) -
      ROW_NUMBER() OVER (PARTITION BY Profit_Day ORDER BY Date) AS grp
   FROM ( ... ) t
) t2
WHERE Profit_Day = 1
GROUP BY grp
ORDER BY Streak_Length DESC;
```

-- The dataset shows a significant drawdown during early 2000, with consecutive negative returns indicating a sustained bearish trend. The stock also exhibits high volatility with sharp recoveries, suggesting it is a high-risk asset suitable for investors with strong risk tolerance.

Date	Close	Daily_Return
2000-01-03	4.471373558	NULL
2000-01-04	4.227483273	-0.05454482427746199
2000-01-05	4.471373558	0.05769160260376978
2000-01-06	4.448146343	-0.005194648735720656
2000-01-07	4.448146343	0
2000-01-10	4.523636341	0.016971113848085737
2000-01-11	4.750109673	0.05006444261386756
2000-01-12	4.5875144	-0.03422979345597101
2000-01-13	4.645586014	0.0126586227173478
2000-01-14	4.767531872	0.026249833203497355
2000-01-18	4.703653812	-0.013398559614285972

-- WHAT IS A RISK MODEL?It answers:

```
-- How risky is this stock?
-- Is the return worth the risk?
-- How stable is performance?
WITH returns AS (
   SELECT
      STR_TO_DATE(Date, '%Y-%m-%d') AS Date,
      Close,

      (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%Y-%m-%d')))
      / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%Y-%m-%d')) AS Daily_Return
```

```
    FROM nke
)

SELECT
    AVG(Daily_Return) AS Avg_Return,
    STDDEV(Daily_Return) AS Volatility,

    AVG(Daily_Return) / STDDEV(Daily_Return) AS Sharpe_Ratio

FROM returns;
```

Avg_Return	Volatility	Sharpe_Ratio
0.0004539078163521557	0.022747468432763998	0.019954212386041864

```
--  “How can we automatically detect MARKET CRASHES?
SELECT
    STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
    Close,

    (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
    / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return

FROM nke;
SELECT *
FROM (
    SELECT
        STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
        Close,

        (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
        / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return

    FROM nke
) t
WHERE Daily_Return < -0.05
ORDER BY Date;
-- “I identified extreme downside events in Nike stock using SQL by filtering daily returns below
-5%.
```

-- The analysis revealed clustering during major economic crises like 2008 and COVID-19, and showed how recovery speed differs across periods.
-- This helps in building risk-aware investment strategies.”

Date	Close	Daily_Return
2000-01-04	4.227483273	-0.05454482427746199
2000-02-08	3.437733412	-0.18231989583313876
2000-02-09	3.135770082	-0.08783791347692785
2000-02-15	2.82799983	-0.05802712946982378
2000-05-03	3.944303036	-0.06353572359088416
2000-05-17	3.845407486	-0.06506308601460413
2000-08-22	3.927074194	-0.10133363877434978
2000-09-11	3.41434288	-0.06389712338244626
2000-09-20	3.536365271	-0.09821441152892361
2000-10-06	3.705598831	-0.07703431644457069
2001-01-19	4.902585983	-0.06465937990486997

-- How often do crashes happen each year?”

```
SELECT
   YEAR(STR_TO_DATE(Date, '%m/%d/%Y')) AS Year,
   COUNT(*) AS Crash_Days
FROM (
   SELECT
      Date,
      (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
      / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return
   FROM nke
) t
WHERE Daily_Return < -0.05
GROUP BY Year
ORDER BY Crash_Days DESC;
```

-- “I analyzed crash frequency per year and found that extreme risk is concentrated during major economic events like 2000 and 2008. This shows that volatility is cyclical and helps investors adjust their strategy based on market conditions.”

Year	Crash_Days
2000	10
2008	10
2022	8
2001	6
2020	5
2009	4
2025	4
2011	3
2024	3
2002	2
2003	2

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```
-- What is the BEST HOLD duration after buying?”
WITH returns AS (
   SELECT
      STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
      Close,

      (Close - LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')))
      / LAG(Close) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Daily_Return,

      LEAD(Close, 30) OVER (ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')) AS Close_30d
   FROM nke
)

SELECT
   AVG((Close_30d - Close)/Close) AS Return_After_Crash_30d
FROM returns
WHERE Daily_Return < -0.05;
```

-- “I analyzed post-crash performance and found that buying Nike after a 5% drop yields an average 2.45% return over 30 days, indicating mean reversion and supporting a short-term recovery trading strategy.”

	Return_After_Crash_3...	
	0.02450071984737249	

```
-- Build complete trading strategy in SQL
WITH base AS (
   SELECT
      STR_TO_DATE(Date, '%m/%d/%Y') AS Date,
      Close
   FROM nke
),

returns AS (
   SELECT
      Date,
      Close,

      (Close - LAG(Close) OVER (ORDER BY Date))
      / LAG(Close) OVER (ORDER BY Date) AS Daily_Return,

      LEAD(Close, 30) OVER (ORDER BY Date) AS Sell_Price
   FROM base
),

trades AS (
   SELECT
      Date AS Buy_Date,
      Close AS Buy_Price,
      Sell_Price,

      (Sell_Price - Close) / Close AS Trade_Return

   FROM returns
   WHERE Daily_Return < -0.05
)
```

```
SELECT
   COUNT(*) AS Total_Trades,

   AVG(Trade_Return) AS Avg_Return,

   SUM(CASE WHEN Trade_Return > 0 THEN 1 ELSE 0 END)
   / COUNT(*) AS Win_Rate,

   SUM(Trade_Return) AS Total_Return

FROM trades;
```

-- “I developed a SQL-based trading strategy that buys Nike stock after significant price drops and evaluates performance using metrics like win rate, average return, and total return. The strategy achieved a 53% win rate and 159% cumulative return, demonstrating how data-driven approaches can identify profitable market inefficiencies.”

Total_Trades	Avg_Return	Win_Rate	Total_Return
66	0.02450071984737249	0.5303	1.5925467900792118

-- “How many BUY and SELL signals are generated?”

```
SELECT
   trade_signal,
   COUNT(*) AS total_signals
FROM (
   SELECT
      STR_TO_DATE(Date, '%m/%d/%Y') AS dt,
      Close,

      CASE
         WHEN Close > AVG(Close) OVER (
            ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')
            ROWS BETWEEN 49 PRECEDING AND CURRENT ROW
         ) THEN 'BUY'
         ELSE 'SELL'
      END AS trade_signal

   FROM nke
) t
GROUP BY trade_signal;
```

-- “I evaluated trading signals not just on next-day returns but by simulating multi-day holding periods, which better reflects real trading behavior.”

trade_signal	total_signals
SELL	2659
BUY	3900

```
-- What if I HOLD for 5 days after BUY?”
WITH signals AS (
    SELECT
        STR_TO_DATE(Date, '%m/%d/%Y') AS dt,
        Close,

        CASE
            WHEN Close > AVG(Close) OVER (
                ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')
                ROWS BETWEEN 49 PRECEDING AND CURRENT ROW
            ) THEN 1
            ELSE 0
        END AS buy_flag,

        LEAD(Close, 5) OVER (
            ORDER BY STR_TO_DATE(Date, '%m/%d/%Y')
        ) AS future_price

    FROM nke
)

SELECT
    COUNT(*) AS total_trades,

    SUM(CASE
        WHEN future_price > Close THEN 1
        ELSE 0
    END) AS profitable_trades,

    SUM(CASE
        WHEN future_price > Close THEN 1
```

```
        ELSE 0
    END) / COUNT(*) AS win_rate,

    AVG((future_price - Close)/Close) AS avg_return

FROM signals
WHERE buy_flag = 1;
```

-- “I developed a moving average-based trading strategy and evaluated its performance using SQL. The model achieved a 54% win rate with an average return of 0.21% per trade over 3900 trades, demonstrating a consistent and statistically significant edge in the market.”

Result Grid Filter Rows: Search Export:

total_trad...	profitable_trad...	win_rate	avg_return
3900	2115	0.5423	0.002101455144565213

-- Is this strategy ACTUALLY better than just holding the stock?”

```
WITH base AS (
    SELECT
        STR_TO_DATE(Date, '%m/%d/%Y') AS dt,
        Close
    FROM nke
),

-- Buy & Hold return
buy_hold AS (
    SELECT
        (MAX(Close) - MIN(Close)) / MIN(Close) AS total_return
    FROM base
),

-- Strategy return
signals AS (
    SELECT
        dt,
        Close,

        CASE
            WHEN Close > AVG(Close) OVER (
                ORDER BY dt
```

```
            ROWS BETWEEN 49 PRECEDING AND CURRENT ROW
        ) THEN 1
        ELSE 0
    END AS buy_flag,

    LEAD(Close, 5) OVER (ORDER BY dt) AS future_price
  FROM base
),

strategy AS (
  SELECT
    AVG((future_price - Close)/Close) * COUNT(*) AS total_return
  FROM signals
  WHERE buy_flag = 1
)

SELECT
  'Buy & Hold' AS method, total_return
FROM buy_hold

UNION ALL

SELECT
  'Trading Strategy', total_return
FROM strategy;
```

-- “While the strategy generated positive returns and a win rate above 50%, it underperformed a buy-and-hold approach due to frequent trading and short holding periods. This indicates that the asset follows a strong long-term trend, and optimizing holding duration or adding trend filters would improve performance.”

method	total_return
Buy & Hold	66.3618507496978
Trading Strategy	8.19567506380433